ConsILR

The Hidden Core of Romanian

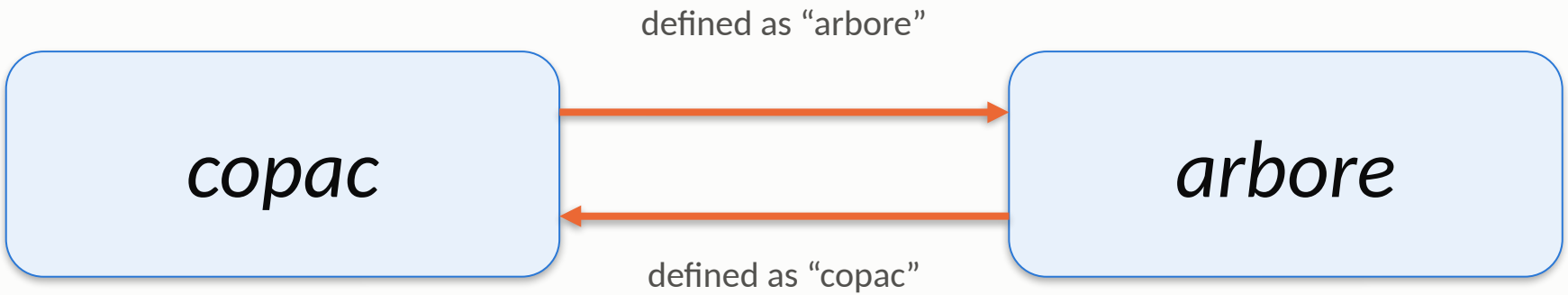
A graph-theoretic analysis of dictionary definitions

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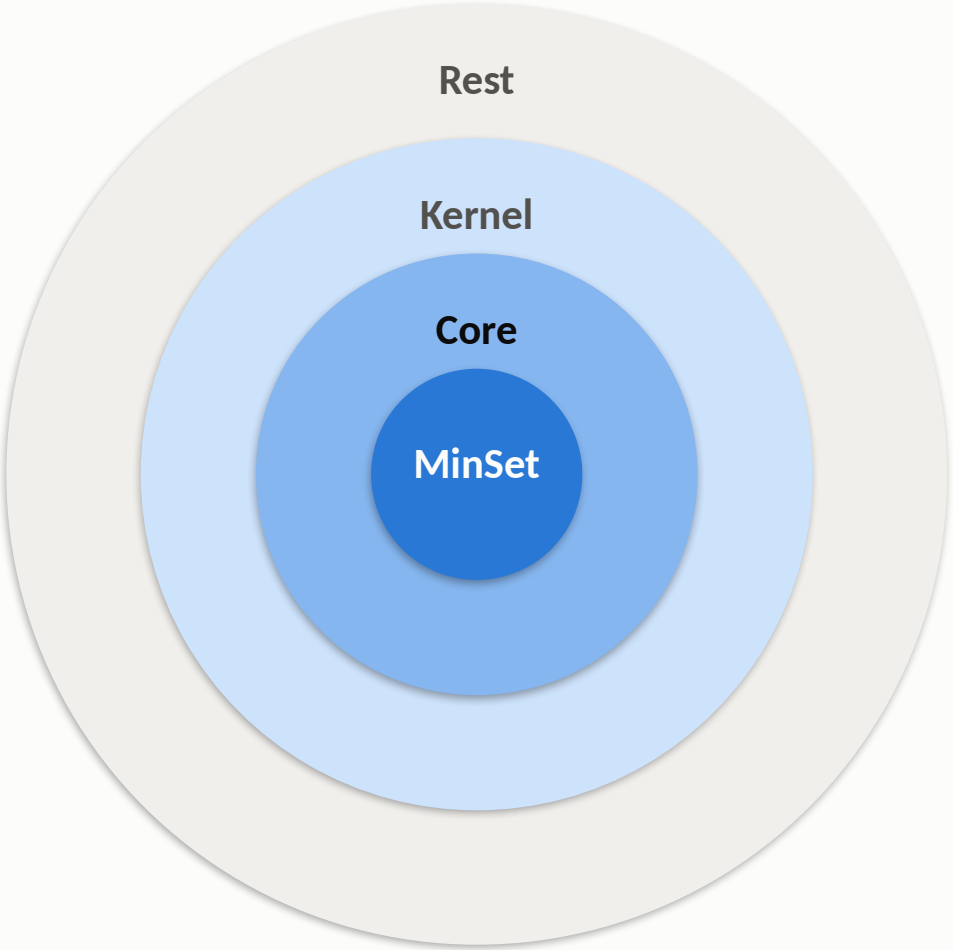
A dictionary is a circle



How many words must you already know before the dictionary can teach you all the others?

Somewhere the circle must be broken by words learned from experience (the symbol grounding problem, Harnad 1990).

Dictionaries as graphs: Kernel, Core, MinSet



Graph: an arrow from every defining word to the word it defines.

Kernel: what remains after repeatedly removing words that define nothing. Everything else can be defined from it.

Core: its largest strongly connected part.

MinSet: the smallest set of words from which the whole dictionary can be learned (an NP-hard problem).

English dictionaries: Kernel 7–12%, MinSet ≈1%.

Romanian linguistics asked a similar question in 1954

Al. Graur, *Încercare asupra fondului principal lexical al limbii romîne* (1954)

1,419 words, in three circles of certainty (964 · 217 · 238), each labelled with its origin.

Criteria: used by the whole people, old and stable, productive, rich in expressions.

His own test: a core word occupies a large space in dictionaries (pp. 24–25).

He called the selection subjective (1957). We treat it as a historical expert hypothesis.

We transcribed the list

1,419 entries · Slavic share 21.49% (exactly as published)

1,416 linked to DEX headwords

Graur left out arbore because it competes with copac, pom, lemn (p. 19).

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Reading the whole DEX, with a language model as an instrument

65,452

definitions, DEX 2nd edition (dexonline, GPL)

63,843

headwords (DEX ’98 completed with DEX ’96 for G–I)

26%

of word forms in definitions are ambiguous

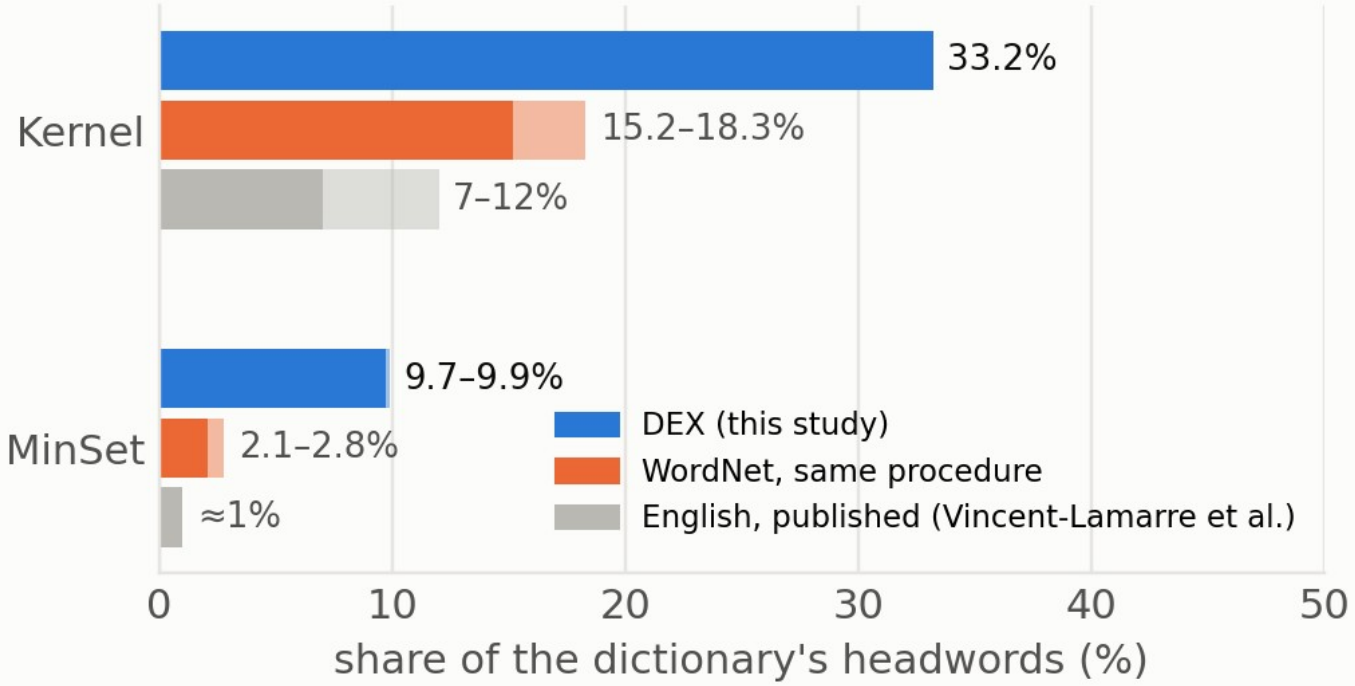
sare → sare ‘salt’ or sări ‘jump’? *fire* → fir ‘thread’, fire ‘nature’ or fi ‘be’?

Claude Haiku 4.5 reads each definition and picks the lemma; a stronger model (Claude Opus 5) checks a random sample: **87% agreement on content words.**

Cost ≈ 23 USD. Every result is repeated without the model (frequency rule, all candidates).

RESULT 1

The DEX has a very large hidden core



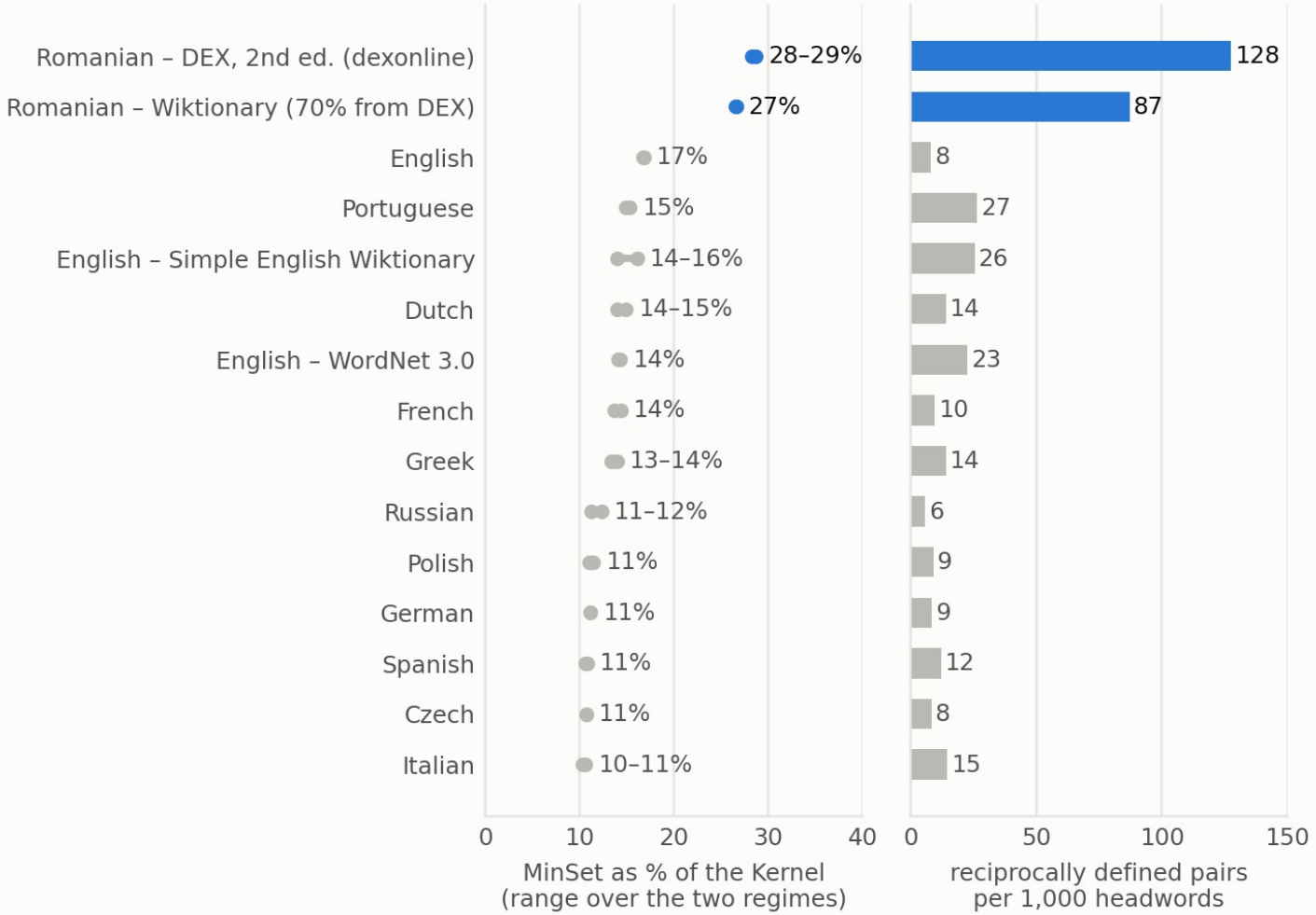
33%

of headwords in the Kernel (21,216 words)

9.7–9.9%

MinSet: 6,193–6,325 words (proven lower and upper bounds)

The DEX stands apart from 11 other languages



28% of the DEX Kernel is needed as MinSet

10–17% in all other dictionaries

Reciprocal pairs: 5–20 times more frequent in the DEX.

Italian, Spanish, Portuguese and French behave like German or Russian: not a Romance property.

Why? Definition by synonym

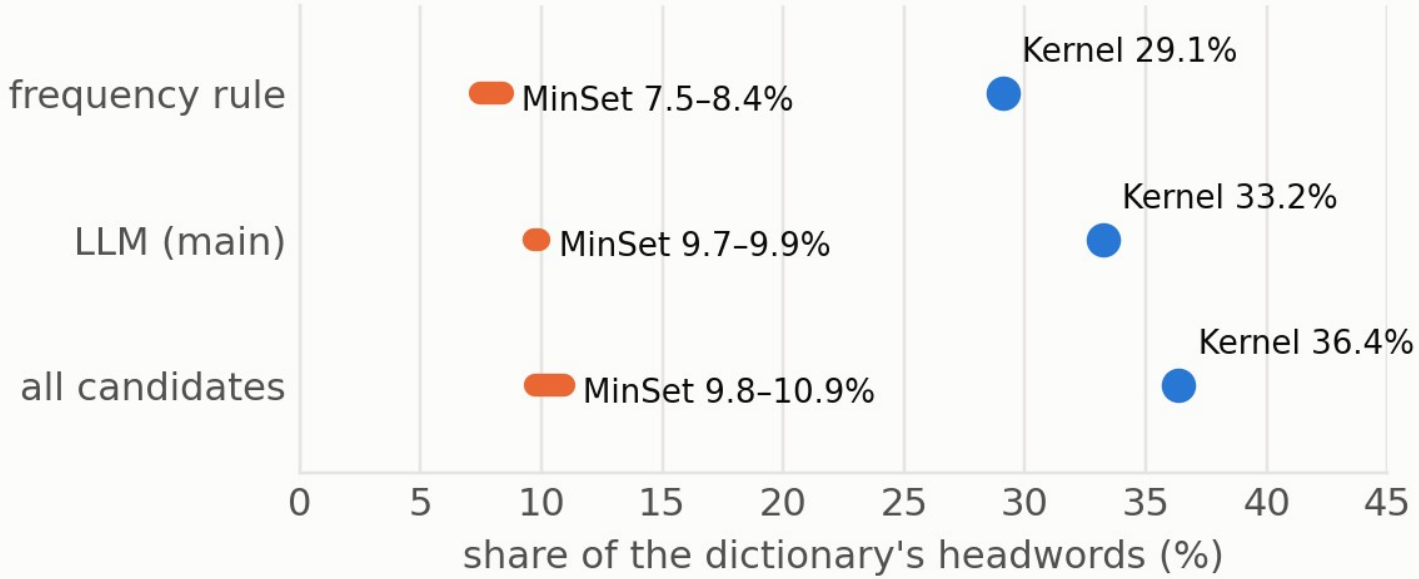
arbore ⇄ *copac*	*amabil* ⇄ *binevoitor*	*primejdie* ⇄ *risc*
balenă ⇄ *chit*	*capelă* ⇄ *paraclis*	*apuca* ⇄ *înșfăca*

8,761 pairs of words define each other (29% share a stem; most others are synonyms).

Each pair forces one word into the MinSet: 98% of the disjoint cycles are such pairs.

English learner dictionaries use a controlled defining vocabulary; the DEX defines by synonym.

The conclusion does not depend on the AI



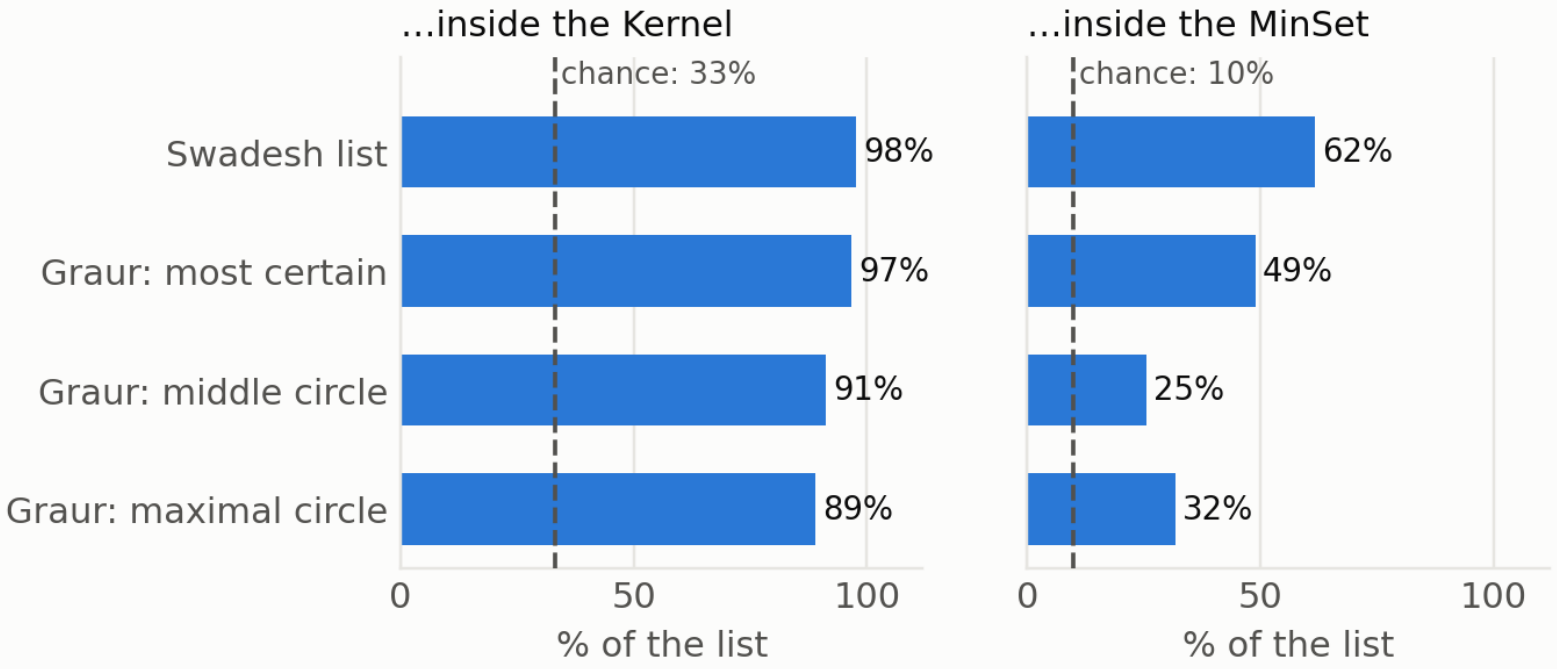
Three ways of resolving ambiguous forms: the most frequent lemma, the language model, all candidates.

Same picture every time: a Kernel of 29–36% and a MinSet of 7.5–10.9%.

Correct lemmas reveal more reciprocal pairs.

RESULT 3

The core is linguistically real



98% of the Swadesh list (about 200 basic meanings used in comparative linguistics) in the Kernel

95% of Graur's words in the Kernel

chance: 33%

Graur's most certain words are the most central: 97% in the Kernel, 49% in the MinSet (wider groups: 25%, 32%).

Graur’s own examples, seen through the graph

word	Graur 1954	graph layer	used in definitions of
arbore	not in the list	MinSet	157 headwords
copac	middle circle	MinSet	84 headwords
pom	most certain	Core	52 headwords
femeie	maximal circle	MinSet	467 headwords
nevastă	not in the list	Core	18 headwords
muiere	maximal circle	Core	7 headwords
cioban	not in the list	MinSet	27 headwords
păcurar	not in the list	MinSet	6 headwords
doagă	middle circle	Core	23 headwords
săgeată	most certain	Core	17 headwords
concert	not in the list	Core	15 headwords
axiomă	not in the list	outside Kernel	2 headwords

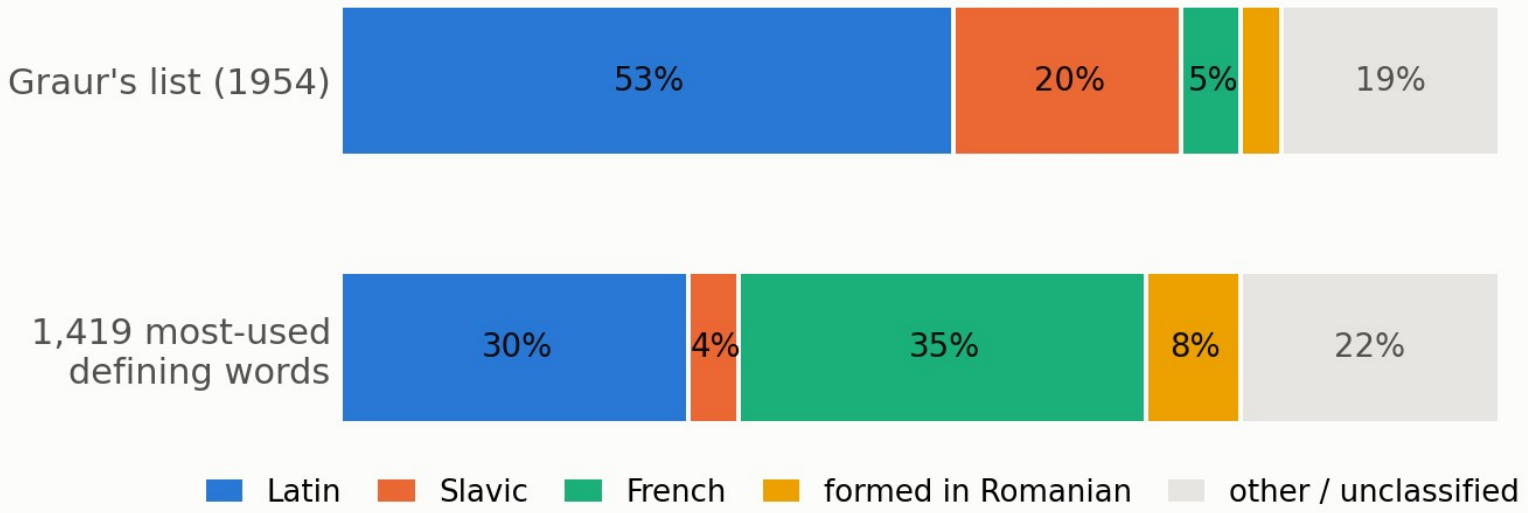
arbore was excluded by Graur because it competes with copac, pom, lemn (p. 19).

Yet the DEX uses arbore in 157 definitions, copac in 84.

The dictionary’s defining preferences and the speakers’ usage Graur described are different things.

RESULT 4

Two cores: inherited words and working words



Graur's core: stable, inherited (Latin + Slavic).

The DEX's working core: learned words: acțiune, rezultat, persoană, substanță…

Only 438 of Graur's words are among the 1,419 most-used defining words.

Outside the Kernel: *arici*, *mânz*, *țap*, *țarc*, *teasc*, *topor*, *prag*… (rural words that define nothing else)

Implications for Romanian lexicography and linguistics

A map of circularity

the list of mutually defining pairs shows where the DEX relies on synonymy; useful for a revised or learner's dictionary.

A test of a classical idea

Graur's list is supported by data he never had, and placed in context.

AI as an instrument

a narrow, checkable task at full scale; reliability measured; conclusions tested without the model.

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Strength of the evidence

Strong	The DEX is much more circular than dictionaries of 11 other languages processed the same way; the cause is reciprocal (synonym) definition.
Strong	Swadesh and Graur words are concentrated in the Kernel ($p < 10^{-200}$).
Good	MinSet size: proven bounds 6,193–6,325, a gap of 2.1%; not computed exactly.
Good	Graur's most certain words are the most central; two marks out of 1,419 uncertain in our transcription.
Indicative	Etymological contrast: part of the vocabulary is unclassified.
Scope	One Romanian dictionary (the Romanian Wiktionary copies it), one edition, first senses only.

Conclusion

A third of the DEX forms a Kernel; about one word in ten must be known in advance to read the whole dictionary.

In the DEX the MinSet takes 28% of the Kernel, against 10–17% in 11 other languages, mainly because the DEX defines words by synonyms.

The Kernel contains almost all Swadesh and Graur words; Graur’s most certain words are the most central.

A second, modern core of learned words does the defining.

We release the graph, its layers, the mutual pairs and the code.

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